Vansh Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.60
Kishinchand Chellaram College	Class XII (Maharashtra State Board Of Secondary and Higher Secondary Education)	2023	80.83%
S. M. Public High School	Class X (Maharashtra State Board Of Secondary and Higher Secondary Education)	2021	79.40%

SCHOLASTIC AND TECHNICAL ACHIEVEMENTS

- **Expert** on **Codeforces** (max rating : **1633**)
- Solved **600+** problems across different platforms

SKILLS

- **Programming Languages:** C++, C, Python, Java, JavaScript, SQL, HTML, CSS, Assembly(RISC-V)
- **Familiar with:** Bash**,** Pandas, Numpy, MatplotLib
- **Tools**: Git/Github, Docker, SolidEdge, PostgreSQL, LaTeX
- **Soft Skills:** Problem Solving, Leadership, Effective Communication, Team Collaboration, Time Management

EXPERIENCE

Software Developer

Gurujal - Rainwater Harvesting Recommendation Platform | *React · Node.js · Express · PostgreSQL · Leaflet* | GitHub

- Developed responsive React interfaces for vendor and admin portals, improving usability through reusable UI components and responsive layouts across multiple workflows.
- Contributed to backend recommendation logic that suggests **rainwater harvesting systems** using rooftop area, rainfall, and soil characteristics, integrating the recommendation pipeline with frontend modules.
- Built **10+ reusable React components** including forms, dashboards, and interactive map views, reducing duplicate code and simplifying feature development across three application portals.
- Designed and integrated REST APIs for recommendation and vendor management modules using Express and PostgreSQL, collaborating through **Git feature** branches and code reviews within the development team.

PROJECTS

xv6 Operating System Enhancements | *C · xv6 · Operating Systems* | GitHub

- Extended the **xv6 kernel** by implementing custom system calls, a System-Call-Aware MLFQ scheduler, and process monitoring utilities to improve scheduling fairness and provide better runtime process insights.
- Implemented **demand paging, lazy allocation,** and **swapping** so the kernel handles memory pressure without crashing.
- Added **FCFS** and **SSTF** disk scheduling along with **RAID 0/1/5** support for swap space, improving I/O management while ensuring reliability and fault tolerance during disk failures.

GraphLib | *C++17 · STL · CMake · GoogleTest* | GitHub

- Built a header-only C++17 graph algorithms library implementing **11 classical graph algorithms**, including BFS/DFS, Dijkstra, Bellman-Ford, Floyd-Warshall, Prim, Kruskal, Tarjan SCC, Kosaraju SCC, bridge detection, and Edmonds-Karp maximum flow, providing a reusable framework for solving diverse graph problems efficiently.
- Designed a fully **templated architecture** supporting any hashable node type (int, string, char) and generic weight types, separating graph storage from algorithms to improve modularity, extensibility, and maintainability.
- Wrote **107 GoogleTest** cases across **6 suites** to stress-test edge cases like self-loops, disconnected components, and negative-weight cycles, and cross-validated results between algorithm pairs (Tarjan vs. Kosaraju, Dijkstra vs. Bellman-Ford) to catch subtle correctness bugs before they'd ever surface in production use.

Custom RISC-V ISA Simulator Extensions | *C++ · RISC-V · Verilog · Vivado · FPGA* | GitHub

- Extended a **RISC-V ISA simulator** with custom cryptography and number-theory instructions by designing instruction encodings and implementing execution logic in **C++**, following the RISC-V ISA specification.
- Worked with a teammate who handled the Verilog/FPGA implementation (tested on a 200MHz Artix-7 board).
- Cross-checked instruction outputs against **C++** and **Python** reference implementations to catch encoding bugs.

Arbitrary Precision Arithmetic Library | *Java · OOP · Ant · Docker* | GitHub

- Wrote a **Java library** for arbitrary-precision integer and floating-point math from scratch - stored numbers as digit lists and implemented add/subtract/multiply/divide directly on them, instead of relying on BigInteger/BigDecimal.
- Packaged the project as a command-line application using **Ant** and **Docker**, enabling reproducible builds, platform-independent execution, and a consistent development workflow across different environments.
- Handled **edge cases** including division by zero, negative numbers, configurable floating-point precision, and invalid inputs, validating results against manual calculations to ensure correctness and numerical accuracy.

RELEVANT COURSES

- **Computer Science:** Discrete Mathematics, Artificial Intelligence, Software Development Fundamentals, Computer Architecture, Data Structures, Algorithms, Theory of Computation, Operating Systems, Database Management Systems
- **Mathematics:** Calculus, Probability and Random Variables, Linear Algebra, Transform Techniques, Statistics

POSITIONS OF RESPONSIBILITY

- **PR and Publicity Head**, Sci-Tech Council IITH 2025-26

- **Core**, RDM, Dramatic and Speaking Arts Club, IITH